

Monitoring male reproductive success in a Japanese black pine clonal seed orchard with RAPD markers

S. Goto, F. Miyahara, and Y. Ide

Abstract: Overall male reproductive success was investigated in a Japanese black pine (*Pinus thunbergii* Parl.) clonal seed orchard consisting of 16 nematode-resistant clones, using random amplified polymorphic DNA (RAPD) markers. We genotyped all 16 of the orchard clones at selected RAPD loci by analyzing diploid needles and 10 haploid megagametophytes from each of them. In addition, the male parent of open-pollinated seedlings was identified by comparing their phenotypes with the genotypes of the orchard clones. In this way the male parents of 559 seedlings of 648 seedlings of progenies from 14 parental (orchard) clones were determined. The level of external pollen contamination in the seed orchard was estimated to be 2.2% for the total population. The effective mean selfing rate observed was 1.5%. Male reproductive success among clones varied widely, from 0.8 to 24.4%, for the total population. Our data suggest that the significant deviation in male reproductive success of individual clones from panmictic ratios may be a problem for the genetic quality in seed orchard crops. To improve the parental balance of seed orchard crops, the use of artificial management techniques such as gibberellin $A_{4/7}$ treatment, supplemental mass pollination, and rearrangement of clonal ramets is recommended in seed orchards.

Résumé : À l'aide de marqueurs d'amplification de l'ADN de position indéterminée (RAPD), les auteurs ont étudié la réussite globale de reproduction des mâles dans un verger à graines clonal de pin noir (*Pinus thunbergii* Parl.) établi au Japon et constitué de 16 clones résistants aux nématodes. Les génotypes des 16 clones du verger ont été déterminés pour chacun des loci choisis de RAPD en analysant des tissus diploïdes d'aiguilles ainsi que 10 mégagamétophytes haploïdes pour chaque clone. De plus, le parent mâle des semis issus de pollinisation libre a été déterminé en comparant les phénotypes des semis aux génotypes des clones du verger. De cette façon, les auteurs ont pu identifier les parents mâles de 559 semis parmi les 648 semis de descendance découlant de 14 parents clonaux du verger. Le degré de contamination pollinique exogène dans le verger à graines s'établissait à 2,2%, pour l'ensemble de la population. En moyenne, le taux effectif d'autogamie était de 1,5%. La réussite de reproduction des mâles variait beaucoup d'un clone à l'autre, avec des valeurs allant de 0,8 à 24,4% pour l'ensemble de la population. Les données supportent l'idée voulant qu'un écart significatif entre la réussite de reproduction des mâles de chaque clone et les ratios panmictiques puisse affecter négativement la qualité génétique des récoltes de graines du verger. Afin de tendre vers une représentation plus équilibrée des parents dans les récoltes de graines du verger, les auteurs suggèrent la mise en place de stratégies de gestion intensive comme le recours aux traitements à l'acide gibbérellique $A_{4/7}$, la pollinisation supplémentaire de masse ou encore, le réarrangement des ramets clonaux au sein des vergers à graines.

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Introduction

As the use of seed orchards for reforestation increases, the genetic quality and composition of orchard crops is a matter of increasing concern among foresters (Szmidt 1987). The genetic efficiency of a seed orchard has been defined as the degree to which the orchard crop reflects the genetic superiority and diversity present in the orchard population (Adams

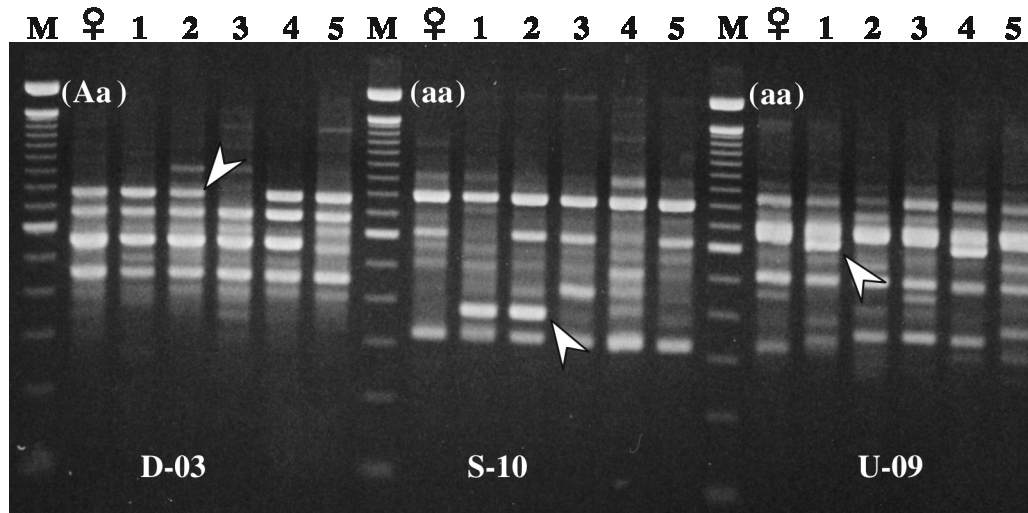
and Joly 1980; El-Kassaby et al. 1984). To obtain full genetic efficiency, all orchard genotypes should contribute equally to the seed orchard crop through their male and female gametes, and pollen contamination and selfing should be negligible (Eriksson et al. 1973). Thus, male reproductive success in fertilization is of particular interest in seed orchards (Wheeler and Jech 1992). Schoen and Stewart (1986) employed multilocus isozyme analysis to identify male parents of progenies from seven clones in a white spruce (*Picea glauca* Moench (Voss)) seed orchard consisting of 33 orchard clones and to evaluate the relationships between male reproductive investment and male reproductive success. Stoehr et al. (1998) used chloroplast simple sequence repeat markers to identify male parents of progenies from six clones in a Douglas-fir (*Pseudotsuga menziesii* (Mirb.) Franco) seed orchard consisting of 20 orchard clones and evaluated pollen contamination, SMP (supplemental mass pollination) management, and selfing. However, knowledge

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Fig. 1. RAPD patterns of a female parent clone (♀) and its progeny obtained by primers D-03, S-10, and U-09. Lanes 1–5 are progeny of the female parent clone (Tsuyazaki(t)-50). Arrows show the RAPD markers used in the study. The putative genotype of the female parent clone is given in parentheses. The following conclusions can be deduced about the male parental genotype: it should be Aa or aa at the D-03 (800 bp) locus for progeny No. 3; AA or Aa for Nos. 1 and 2 at the S-10 (390 bp) locus; Aa or aa for Nos. 3, 4, and 5 at the S-10 locus; AA or Aa at the U-09 (610 bp) locus for Nos. 1 and 4, and Aa or aa for Nos. 2, 3, and 5 at the U-09 locus. M, 100-bp size marker (Gibco BRL).



of the male reproductive success of the individual clones involved is generally limited in seed orchard studies.

In a previous study, we analyzed random amplified polymorphic DNA (RAPD) markers (Welsh and McClelland 1990; Williams et al. 1990) to identify the male parents of a half-sib progeny in a clonal Japanese black pine (*Pinus thunbergii* Parl.) seed orchard consisting of 16 pinewood-nematode-resistant clones (S. Goto, F. Miyahara, and Y. Ide, unpublished). All of the 16 orchard clones were genotyped at 28 RAPD loci by analyzing both diploid needles and 10 haploid megagametophytes of each orchard clone. The male parents of 82 of 85 seedlings produced from a single clone were identified by comparing the phenotypes of each seedling with the genotypes of orchard clones at 28 RAPD loci. In this study, we applied this method to the other progenies in the seed orchard, to obtain a more detailed picture of male reproductive success in seedlings of these orchard crops.

Materials and methods

Seed orchard

The orchard used in this study is a clonal Japanese black pine seed orchard located in Ogoori, Fukuoka Prefecture, Japan. The orchard (0.5 ha in size) consists of 16 nematode-resistant clones selected for use in a project on nematode resistance of Japanese black and red pines (Fujimoto et al. 1989). Clonal ramets were multiplied by grafting, planted 5 × 5 m apart and systematically distributed in the orchard (Giertych 1965).

Plant materials

Needles and haploid megagametophytes dissected from 10 open-pollinated seeds per clone were used for marker selection and genotyping at the selected RAPD loci. Open-pollinated seedlings ($n = 38$ –50) were collected from 14 parental (orchard) clones to monitor male reproductive success

of the clones in the seed orchard. For this purpose, cones were collected from a single ramet of each of 14 clones from the seed orchard in October 1997 and seeds extracted from them were sown in April 1998. Needles were collected from each of the seedlings in June 1999 and were used for identifying the male parent.

DNA extraction and polymerase chain reaction (PCR) amplification

For each sample, DNA was extracted from about 150 mg needles and all of the megagametophytes using a modified cetyltrimethylammonium bromide (CTAB) method (Shiraishi and Watanabe 1995). Extracted DNA was then purified using an ELU-QUIK™ DNA purification kit (Schleicher & Schuell Co.). Purified DNA was used as template DNA for RAPD analysis. RAPD reactions were performed in 20-μL volumes, containing 10 ng of template DNA, 2.0 μL of 10 × Stoffel buffer (100 mM Tris-HCl, 100 mM KCl, pH 8.3), 0.20 mM of each dNTP, 0.25 μM of RAPD primer, 3.0 mM MgCl₂, and 1 unit of AmpliTaq DNA polymerase Stoffel Fragment (Perkin-Elmer Cetus). DNA was amplified in a TaKaRa PCR thermal cycler MP (TP-3000) using the following program: 1 min at 94°C; 45 cycles of 10 s at 94°C, 30 s at 36°C, 60 s at 72°C; and a final extension of 7 min at 72°C. After amplification, the products were separated on 2.0% agarose gels by electrophoresis and were stained with ethidium bromide. The gels were then photographed on an ultraviolet transilluminator.

Marker selection and genotyping the orchard clones

To identify sufficient RAPD loci for male-parent determination, RAPD analysis was carried out on needles of 16 orchard clones. Markers that were polymorphic among orchard clones and reproducible were selected from 320 RAPD primers (Operon Technology). The reproducibility of the markers was confirmed by comparing results from replicates

Table 1. Progenies used for the study and the number of seedlings for which the male parent was identified.

Female parent of progeny	No. of seedlings	No. of RAPD loci	Male parent identified	
			No.*	%
Ei(t)-425	46	28	40	87.0
Misaki(t)-90	49	28	41	83.7
Mitoyo(t)-103	42	31	40	95.2
Namikata(t)-37	49	28	44	89.8
Namikata(t)-73	47	31	34	72.3
Obama(t)-30	48	25	43	89.6
Ooita(t)-8	45	26	37	82.2
Ooseto(t)-12	48	31	37	77.1
Sendai(t)-290	38	28	34	89.5
Shima(t)-64	47	29	43	91.5
Tosashimizu(t)-63	45	28	42	93.3
Tsuyazaki(t)-50	46	28	39	84.8
Yasu(t)-37	49	27	42	85.7
Yoshida(t)-2	49	29	43	87.8
Total	648		559	86.3

*Number of seedlings identified the male parent in the progeny.

of 3–5 PCR analyses. The absence of markers in any clone showed it to be a recessive homozygote (aa) at the corresponding loci. Megagametophytes dissected from 10 open-pollinated seeds per clone were used for distinguishing whether other clones were dominant homozygotes (AA) or heterozygous (Aa) at each RAPD locus. If the marker was present in all of the megagametophytes, the genotype of the clone was identified as a dominant homozygote. If the marker was absent from more than one haploid megagametophyte, it was identified as heterozygous. For 10 megagametophytes, the probability of incorrectly identifying a clone's genotype at any one locus is less than 0.002 using this method (Morris and Spieth 1978).

Investigating the inheritance of RAPD markers

Thirty-two megagametophytes from the orchard clone Namikata(t)-73 were used to study the inheritance of RAPD markers for this study. The dominance and Mendelian traits of RAPD markers were investigated by segregation analysis of these megagametophytes. The agreement of the observed segregation ratios of the RAPD markers with those expected for a dominant Mendelian trait was calculated using χ^2 tests.

Identifying the male parent of the seedlings

The male parent of each open-pollinated seedling was identified by comparing their phenotypes with genotypes of the orchard clones at each RAPD locus. In cases where the female parent of the diagnostic progeny had a recessive homozygous (aa) genotype, the following procedure was used for identifying the male parent. If a marker was present in the seedling, the recessive homozygote (aa) clones at the locus could be excluded as candidates for the male parent. Conversely, if the marker was absent in the seedling, the dominant homozygote (AA) clones could be excluded from the male parent candidates. In cases where the female parent genotypes of the progeny were heterozygous (Aa), the domi-

Table 2. Segregation of RAPD fragments in 32 megagametophytes from the parent clone Namikata(t)-73 and χ^2 analysis of the goodness of fit to a 1:1 ratio.

Primer	bp	Observed*		χ^2	p
		+	–		
A-09	510	18	14	0.25	>0.60
A-11	850	19	13	0.57	>0.40
A-16	620	19	13	0.57	>0.40
A-17	450	16	16	0.00	>0.90
C-09	670	19	13	0.57	>0.40
D-03	700	13	19	0.57	>0.40
F-03	720	16	16	0.00	>0.90
G-17	590	14	18	0.25	>0.60
I-10	530	21	11	1.60	>0.20
I-10	750	11	21	1.60	>0.20
M-17	610	14	18	0.25	>0.60
U-13	580	15	17	0.06	>0.70
V-01	340	14	18	0.25	>0.60
V-01	400	18	14	0.25	>0.60
V-04	850	15	17	0.06	>0.70
V-17	780	17	15	0.06	>0.70

*+, fragment present; –, fragment absent.

nant homozygote (AA) clones could be excluded from the male parent candidates only if the marker was absent in the seedling (Fig. 1). Pollen contamination was also identified through a similar procedure. For example, as progeny No. 1 in Fig. 1 possesses the S-10 (390 bp) fragment, the male parent genotype of this progeny at this locus should be a dominant homozygote or heterozygote for this marker. Thus, the only possible male parent candidates for progeny No. 1 among the orchard clones are Namikata(t)-73, Misaki(t)-90, and Ooseto(t)-12. However, progeny No. 1 also possesses the U-09 (610 bp) fragment, whereas these three candidate clones at this RAPD locus were all recessive homozygotes. Therefore, the male parentage of No. 1 was attributed to pollen contamination from a tree outside the orchard.

One of the 16 orchard clones, Obama(t)-24, is known to be a natural hybrid (*Pinus densi-thunbergii* Uyeki) clone whose female and parents are, respectively, a Japanese black pine and a Japanese red pine (*Pinus densiflora* Sieb. et Zucc.) (Watanabe et al. 1996). In most conifers (including Japanese black and red pines; Watanabe et al. 1996), chloroplast DNA (cpDNA) is inherited from the pollen donor (Wagner 1992). Therefore, Obama(t)-24 possesses cpDNA originating from Japanese red pine. Shiraishi and Watanabe (1995) investigated cpDNA variations in the *rbcL* gene between Japanese black and red pines, and demonstrated a method of distinguishing between the cpDNAs of the two species by PCR – restriction fragment length polymorphism analysis. We used this method to identify the male parent of Obama(t)-24 from the male parent candidates in each progeny.

Male reproductive success (MRS) is defined as the proportion of the seed orchard crop pollinated by the respective clone in a seed orchard. The value of $MRS_{(ij)}$ ranged from $1/N_{(i)} \times SMIN_{(ij)}$ to $1/N_{(i)} \times SMAX_{(ij)}$, where $MRS_{(ij)}$ represents the male reproductive success of the *j*th parental (orchard) clone for the *i*th progeny. $SMIN_{(ij)}$ is defined as the minimum number of seedlings pollinated by the *j*th clone for

Table 3. Male reproductive success in open-pollinated seedlings collected from 14 orchard clones in a Japanese black pine seed orchard.

Female parent of the progeny	N	Male parent							
		Ei(t)-425	Misaki(t)-90	Mitoyo(t)-103	Namikata(t)-37	Namikata(t)-73	Obama(t)-30	Ooita(t)-8	Ooseto(t)-12
Ei(t)-425	46	0.000±0.000*	0.109±0.000	0.000±0.000	0.141±0.011	0.087±0.022	0.000±0.000	0.043±0.043	0.087±0.000
Misaki(t)-90	49	0.010±0.010	0.000±0.000*	0.051±0.051	0.122±0.020	0.041±0.020	0.041±0.020	0.031±0.010	0.092±0.031
Mitoyo(t)-103	42	0.000±0.000	0.083±0.012	0.000±0.000*	0.202±0.012	0.024±0.000	0.000±0.000	0.012±0.012	0.155±0.012
Namikata(t)-37	49	0.071±0.010	0.031±0.010	0.000±0.000	0.041±0.000*	0.020±0.000	0.020±0.000	0.000±0.000	0.173±0.031
Namikata(t)-73	47	0.043±0.043	0.149±0.021	0.043±0.043	0.074±0.074	0.011±0.011*	0.053±0.053	0.085±0.043	0.160±0.032
Obama(t)-30	48	0.000±0.000	0.104±0.000	0.000±0.000	0.125±0.042	0.094±0.052	0.010±0.010*	0.031±0.010	0.198±0.010
Ooita(t)-8	45	0.011±0.011	0.044±0.022	0.022±0.022	0.178±0.000	0.067±0.044	0.000±0.000	0.000±0.000*	0.633±0.078
Ooseto(t)-12	48	0.021±0.021	0.063±0.000	0.042±0.042	0.156±0.031	0.115±0.052	0.042±0.000	0.083±0.042	0.052±0.052*
Sendai(t)-290	38	0.053±0.026	0.263±0.000	0.026±0.026	0.211±0.026	0.000±0.000	0.026±0.000	0.026±0.000	0.211±0.000
Shima(t)-64	47	0.021±0.000	0.064±0.000	0.011±0.011	0.032±0.011	0.085±0.000	0.000±0.000	0.032±0.011	0.000±0.000
Tosashimizu(t)-63	45	0.044±0.000	0.022±0.000	0.000±0.000	0.089±0.000	0.100±0.033	0.022±0.000	0.022±0.000	0.300±0.033
Tsuyazaki(t)-50	46	0.054±0.033	0.207±0.011	0.043±0.043	0.098±0.054	0.174±0.000	0.033±0.033	0.054±0.054	0.065±0.000
Yasu(t)-37	49	0.010±0.010	0.143±0.000	0.020±0.020	0.102±0.041	0.020±0.000	0.000±0.000	0.031±0.010	0.020±0.000
Yoshida(t)-2	49	0.020±0.000	0.163±0.000	0.010±0.010	0.133±0.010	0.000±0.000	0.000±0.000	0.051±0.010	0.184±0.000
Total population	648	0.025±0.012	0.101±0.005	0.019±0.019	0.120±0.024	0.060±0.017	0.018±0.008	0.036±0.018	0.164±0.020

Note: Male reproductive success of Obama(t)-24 was determined by PCR-RFLP analysis of the *rbcL* gene. Male reproductive success of the *j*th clone for the *i*th progeny.

*Effective selfing.

the *i*th progeny. $SMAX_{(ij)}$ is the maximum number of seedlings that could be pollinated by the *j*th clone for the *i*th progeny and $N_{(i)}$ is the number of seedlings in the *i*th progeny. In this study, $MRS_{(ij)}$ is given by the following formula:

$$MRS_{(ij)} = \frac{(SMAX_{(ij)} + SMIN_{(ij)}) \pm (SMAX_{(ij)} - SMIN_{(ij)})}{2N_{(i)}}$$

Results and discussion

Following the marker evaluation, 17 markers were selected for this study. The genotypes of the 16 orchard clones at these loci were determined by analyzing needles and 10 megagametophytes from each of the clones. In this exercise we genotypically characterized 16 orchard clones at 45 RAPD loci: 28 identified by S. Goto, F. Miyahara, and Y. Ide (unpublished data), and 17 corresponding to markers developed in this study. We then chose 25–31 of the 45 loci at which the female parent of the diagnostic progeny was a recessive homozygote (aa) or heterozygote (Aa), to identify the male parents (Table 1). We also investigated the segregation of RAPD fragments at 45 RAPD loci in 32 megagametophytes of Namikata(t)-73. For dominant homozygous RAPD loci, all megagametophytes will possess the corresponding fragments, while for recessive homozygous RAPD loci, no megagametophytes will possess them, and at the heterozygote RAPD loci, segregation of RAPD fragments generally shows a good fit to Mendelian expectations (Table 2). Therefore, the RAPD markers were shown to display dominant–recessive behavior and Mendelian traits in these Japanese black pines, and our results corroborate previous reports obtained in other conifers (Nelson et al. 1993; Lu et al. 1995).

We can identify the male parent for a total of 559 (86.3%) of the 648 seedlings investigated in this study. The discrimination rate ranged from 72.3 to 95.2% depending on the progeny (Table 1). In this study, we applied RAPD markers

to identify the male parents of progenies from 14 clones in a Japanese black pine seed orchard consisting of 16 orchard clones, and we show the male reproductive success of the 16 clones in each investigated progeny in Table 3. It is noteworthy that an almost complete picture of male reproductive success in the seed orchard was obtained in this study.

The male reproductive success of individual clones varied widely, from 0.8 to 24.4%, for the total population in this study (Table 3). Less reproductively successful clones such as Shima(t)-64, Tosashimizu(t)-63, and Yoshida(t)-2 made hardly any contribution as male parents to any of the progenies, but more successful clones such as Tsuyazaki(t)-50, Ooseto(t)-12, Namikata(t)-37, and Sendai(t)-290 dominated as male parents for almost all of the progenies. Assuming panmixis, one would expect every orchard clone to be represented as the male parent at a frequency of 6.25% (1/16) in each progeny. However, in most progenies only two or three clones dominated, acting as male parents for over 50% of the seedlings. The dominance was especially strong in progenies of Ooita(t)-8, Yasu(t)-37, and Shima(t)-64, for which only one clone was the male parent in over 50% of cases. Compared with previous reports (Schoen and Stewart 1986; Stoehr et al. 1998), male parental balance in this seed orchard was seriously biased.

Previous reports have implied that the reproductive success of male parents is significantly correlated to their pollen production (Schoen and Stewart 1986; Burczyk and Prat 1997). We also found that more reproductively successful clones tended to shed their pollen heavily in the mating season (data not shown), and flowering phenology is known to be one of the most important factors in outcrossing patterns in seed orchards (El-Kassaby et al. 1988; Erickson and Adams 1989; Burczyk and Prat 1997). Obama(t)-24, one of the orchard clones in this seed orchard, is known to be a heavy pollen producer (Toda et al. 1993), but the male reproductive success of this clone was only 3.1% for the total population. This value was very similar to the result obtained by Miyahara et al. (1998). The observed values of Obama(t)-24

Sendai(t)-290	Shima(t)-64	Tosashimizu(t)-63	Tsuyazaki(t)-50	Yasu(t)-37	Yoshida(t)-2	Tanabe(t)-54	Obama(t)-24	Contamination
0.087±0.000	0.000±0.000	0.000±0.000	0.380±0.033	0.011±0.011	0.033±0.011	0.000±0.000	0.022	0.022
0.061±0.020	0.020±0.020	0.000±0.000	0.439±0.031	0.031±0.010	0.000±0.000	0.020±0.000	0.061	0.041
0.024±0.000	0.000±0.000	0.000±0.000	0.393±0.012	0.024±0.000	0.000±0.000	0.024±0.000	0.048	0.024
0.214±0.010	0.010±0.010	0.000±0.000	0.245±0.041	0.092±0.010	0.000±0.000	0.020±0.000	0.041	0.041
0.213±0.021	0.032±0.032	0.011±0.011	0.117±0.074	0.096±0.011	0.011±0.011	0.085±0.000	0.000	0.021
0.167±0.000	0.031±0.010	0.000±0.000	0.125±0.000	0.104±0.000	0.031±0.010	0.022±0.022	0.000	0.021
0.000±0.000	0.000±0.000	0.022±0.022	0.033±0.011	0.000±0.000	0.000±0.000	0.000±0.000	0.000	0.022
0.167±0.063	0.000±0.000	0.042±0.000	0.156±0.052	0.198±0.052	0.010±0.010	0.000±0.000	0.021	0.021
0.026±0.026*	0.000±0.000	0.000±0.000	0.132±0.026	0.053±0.026	0.039±0.039	0.000±0.000	0.026	0.000
0.128±0.021	0.000±0.000*	0.000±0.000	0.543±0.011	0.043±0.021	0.011±0.011	0.021±0.000	0.021	0.000
0.244±0.022	0.000±0.000	0.000±0.000*	0.044±0.000	0.000±0.000	0.022±0.000	0.000±0.000	0.067	0.000
0.196±0.043	0.011±0.011	0.065±0.000	0.043±0.043*	0.043±0.043	0.043±0.043	0.043±0.000	0.065	0.022
0.020±0.020	0.010±0.010	0.000±0.000	0.551±0.020	0.020±0.020*	0.010±0.010	0.000±0.000	0.020	0.041
0.143±0.020	0.000±0.000	0.000±0.000	0.173±0.010	0.020±0.020	0.010±0.010*	0.000±0.000	0.041	0.020
0.123±0.019	0.008±0.007	0.010±0.002	0.244±0.026	0.053±0.016	0.015±0.011	0.017±0.002	0.031	0.022

progeny: $MRS_{(ij)} = 1/(2N_{(ij)}) \times [(SMAX_{(ij)} + SMIN_{(ij)}) \pm (SMAX_{(ij)} - SMIN_{(ij)})]$. See Materials and methods. N , number of seedlings in the progeny investigated.

were less than half of the value expected in a panmictic situation. However, the male flowering period of Obama(t)-24 is considerably later than the peak time of female receptivity for most of the other clones (Toda et al. 1994). This lack of flowering phenology overlap with other clones may inhibit the male reproductive success of Obama(t)-24.

We also noted variations in the male reproductive success of individual clones among progenies. For example, that of Tsuyazaki(t)-50 varied from 3.3 to 55.1% depending on the progeny. We found that this variation was correlated with the minimum distance between the male and female parent clones. The highest value of male reproductive success was observed in the progeny where the female parent clone was located closest to Tsuyazaki(t)-50. Similarly, in an isozyme investigation of the pollination pattern in Scots pine (*Pinus sylvestris* L.) seed orchards, Shen et al. (1981) showed that 31% of the fertilizing pollen received by a neighboring tree came from the marker ramet. However, Erickson and Adams (1989) found that the dispersal of rare markers was strongly associated with distance between mating individuals only when floral synchrony among the clones was high, and Toda et al. (1994) showed that the flowering phenology of the orchard clones we studied, except for Obama(t)-24, was almost the same. Therefore, our results corroborate those of previous reports.

The level of external pollen contamination in the seed orchard was estimated to be 2.2% for the total population. In contrast, pollen contamination rates reported in conifer seed orchards have often been over 30% (see Di-Giovanni and Kevan (1991) and Wheeler and Jech (1992) for reviews). Miyahara et al. (1998) reported that no pines existed within 500 m of the seed orchard used in this study. So, we suggest that the absence of nearby natural populations of Japanese black pines may restrict external pollen contamination in this case. In this seed orchard, the observed value of effective selfing was 1.5% for the total population. As self-pollination in conifers affects embryo development, seed germination, and seedling mortality at early stages, the ef-

fective selfing rates reported in conifer seed orchards are generally low (<10%: see Adams and Joly 1980; Shaw and Allard 1982; Ritland and El-Kassaby 1985; Stoehr et al. 1998). Our results show that effective selfing can be ignored in the management of this seed orchard.

Our data also suggest that the significant deviation in male reproductive success of individual clones from panmictic ratios may be a greater problem for the genetic quality in seed orchard crops than either pollen contamination or selfing. In most progenies, very few clones dominated as male parents. Therefore, the full genetic superiority and diversity of orchard clones cannot be inherited by the next generation. To improve the parental balance, artificial management techniques should be implemented for this seed orchard, such as use of the gibberellin $A_{4/7}$ treatment developed to enhance the flowering of male strobili of Pinaceae by Bonnet-Masimbert (1987). Toda et al. (1993) reported that the number of male strobili was seven times greater than in the control when gibberellin $A_{4/7}$ treatment was applied to Japanese black pines. Recently, SMP management (Askew 1992) was also used to improve the parental balance in seed orchards (El-Kassaby et al. 1984). This approach should also be effective in this seed orchard. In our study, we found the level of reproductive success was high for prospective male parents nearest to neighboring female parents. However, clonal ramets are systematically distributed in this seed orchard, so all the ramets of a certain clone were surrounded by the same combination of clones throughout the seed orchard. This arrangement may prevent optimal randomization of male reproductive success of each orchard clone. Thus, we recommend a random layout of ramets for future seed orchards.

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